

Package ‘math3150package’

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Title Package for MATH 3150 - Applied Statistics at SUU

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Description This package contains functions and data files needed for MATH 3150 - Applied Statistics taught at Southern Utah University.

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Author Rick Brown [aut, cre]

Maintainer Rick Brown <richardbrown1@suu.edu>

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baseball	<i>baseball Data Set</i>
----------	--------------------------

Description

This data set contains information for 337 baseball players.

Usage

```
baseball
```

Format

A tibble with 337 rows and 28 variables:

- salary** The salary in \$1000s.
- average** Batting average of the player.
- obp** On base percentage of the player
- runs** Number of runs scored.
- hits** Number of hits in total.
- doubles** Number of doubles hit.
- triples** Number of triples hit.
- homeruns** Number of homeruns hit.
- rbis** Number of runs batted in.
- walks** Number of times walked.
- sos** Number of strikeouts.
- sbs** Number of stolen bases.
- errors** Number of errors committed.
- freeagent** Factor indicating whether the player is a free agent or is eligible for free agency.
- arbitration** Factor indicating whether the player has arbitration or is eligible for arbitration.
- runsperso** Number of runs per strikeout (runs/sos).
- hitsperso** Number of hits per strikeout (hits/sos).

hrsperso Number of homeruns per strikeout (homeruns/sos).
rbisperso Number of rbis per strikeout (rbis/sos).
walksperso Number of walks per strikeout (walks/sos).
obppererror On base percentage per error (obp/errors).
runsperror Number of runs scored per error (runs/errors).
hitspererror Number of hits per error (hits/errors).
hrsperror Number of homeruns per error (homeruns/errors).
sospererror Number of strikeouts per error (sos/errors).
sbsobp Number of stolen bases times on base percentage (sbsobp). \itemsbsrunsNumber of stolen bases times number of runs scored (sbsruns).
sbshits Number of stolen bases times number of hits (sbs*hits).

bf_test	<i>Perform the Brown-Forsythe Test for Equality of Variance</i>
---------	---

Description

When given a model, this will break up into groups of equal size with a default of 2 perform a Brown Forsythe test. By default, the test will be performed with the jackknife residuals.

Usage

```
bf_test(model, num_groups = 2, resid_type = "jackknife", plot_graph = TRUE)
```

Arguments

model	A model of type lm or glm.
num_groups	The number of groups to be used. The default is 2.
resid_type	The type of residuals. The default is "jackknife", but supports "raw", "standard", and "pearson".
plot_graph	Whether to return a plot or not. The default is TRUE.

Value

This function returns an ANOVA table and, if requested a plot of the residuals

Examples

```
mod <- lm(mpg ~ disp, data = mtcars)
bf_test(mod)
```

 births

births Data Set

Description

Data from 1995-1997 for a study that examined pregnancies that resulted in the birth of twins. Births were classified as preterm with intervention (induced labor or cesarean), preterm without procedures, or term/post-term. Researchers also classified the pregnancies by the level of prenatal medical care the mother received (inadequate, adequate, or intensive). The data set consists of 278 cases (rows) with two columns indicating the level of prenatal care and type of birth for each set of twins.

Usage

```
births
```

Format

A tibble with 278 rows and 2 variables:

prenatal A factor indicating the prenatal care the mother received: Adequate, Inadequate, or Intensive.

type A factor indicating the classification of the birth: "Preterm (induced or cesarean)", "Preterm (without procedures)", and "Term or post-term"

 bootstrapping

Outputs and restyles an object from the boot() function from the boot package.

Description

This function relies on the [boot](#) function. It generates R bootstrap replicates of a statistic applied to data. Both parametric and nonparametric resampling are possible. For the nonparametric bootstrap, possible resampling methods are the ordinary bootstrap, the balanced bootstrap, antithetic resampling, and permutation. For nonparametric multi-sample problems stratified resampling is used: this is specified by including a vector of strata in the call to boot. Importance resampling weights may be specified.

Usage

```
bootstrapping(data, statistic, R, ...)
```

Arguments

data The data as a vector, matrix or data frame. If it is a matrix or data frame then each row is considered as one multivariate observation.

statistic	A function which when applied to data returns a vector containing the statistic(s) of interest. When <code>sim = "parametric"</code> , the first argument to <code>statistic</code> must be the data. For each replicate a simulated dataset returned by <code>ran.gen</code> will be passed. In all other cases <code>statistic</code> must take at least two arguments. The first argument passed will always be the original data. The second will be a vector of indices, frequencies or weights which define the bootstrap sample. Further, if predictions are required, then a third argument is required which would be a vector of the random indices used to generate the bootstrap predictions. Any further arguments can be passed to <code>statistic</code> through the <code>...</code> argument.
R	The number of bootstrap replicates. Usually this will be a single positive integer. For importance resampling, some resamples may use one set of weights and others use a different set of weights. In this case <code>R</code> would be a vector of integers where each component gives the number of resamples from each of the rows of weights.
...	See more documentation in the boot function.

Value

This function returns an object of class `boot3150` that works much like an object of class `boot` from the `boot` package.

Examples

```
med <- function(x, i){median(x[i])}
sample_data <- rnorm(100)
boot_out <- bootstrapping(sample_data, med, 1000)
boot_out
```

boot_ci	<i>Outputs and restyles an object from the <code>boot.ci()</code> function from the <code>boot</code> package.</i>
---------	--

Description

This function relies on the [boot.ci](#) function. This function generates 5 different types of equi-tailed two-sided nonparametric confidence intervals. These are the first order normal approximation, the basic bootstrap interval, the studentized bootstrap interval, the bootstrap percentile interval, and the adjusted bootstrap percentile (BCa) interval. All or a subset of these intervals can be generated.

Usage

```
boot_ci(boot_out, conf = 0.95, type = "all", ...)
```

Arguments

boot_out	An object of class "boot" or "boot3150" containing the output of a bootstrap calculation.
conf	A scalar or vector containing the confidence level(s) of the required interval(s).
type	A vector of character strings representing the type of intervals required. The value should be any subset of the values <code>c("norm", "basic", "stud", "perc", "bca")</code> or simply "all" which will compute all five types of intervals.
...	See more documentation in the boot.ci function.

Value

This function returns an object of class `boot3150ci` that works much like an object of class `bootci` from the `boot` package.

Examples

```
med <- function(x, i){median(x[i])}
sample_data <- rnorm(100)
boot_out <- bootstrapping(sample_data, med, 1000)
bootci_out <- boot_ci(boot_out)
```

<code>cars99</code>	<i>cars99 Data Set</i>
---------------------	------------------------

Description

This data set contains information on 109 vehicles from 1999.

Usage

```
cars99
```

Format

A tibble with 109 rows and 11 variables:

Model The vehicle model name.

CityMPG The miles per gallon (MPG) for the vehicle in the city.

HwyMPG The miles per gallon (MPG) for the vehicle on the highway

FuelCap The fuel capacity of the vehicle in gallons.

Weight The weight of the vehicle in lbs.

FrontWt The front weight of the vehicle.

Accel0_30 The time it takes, in seconds, for the vehicle to accelerate from 0 to 30 mph.

Accel0_60 The time it takes, in seconds, for the vehicle to accelerate from 0 to 60 mph.

QtrMile The time it takes, in seconds, for the vehicle to travel a quarter of a mile.

ci_capture	<i>Computes many confidence intervals and returns how many capture the true mean</i>
------------	--

Description

This function reads in a given sample size, mean μ , standard deviation σ , confidence coefficient, confidence interval type ("z" or "t"), and the number of simulated samples desired, and returns a count of how many of the corresponding confidence intervals captured the true mean μ .

Usage

```
ci_capture(n, mu, sigma, conf_level = 95, ci_type, n_ints, plot_graph = TRUE)
```

Arguments

n	Sample size of each sample.
mu	The true population mean.
conf_level	The confidence level. By default, this is 95 for 95% intervals.
ci_type	The type of confidence interval to create. "z" for a z-interval and "t" for a t-interval.
n_ints	The number of intervals to create.
plot_graph	Whether to return a plot or not. The default is TRUE.

Value

This function returns a graph of all the intervals, if requested, and the number of intervals that contained the true mean.

Examples

```
ci_capture(10, 10, 5, 95, "t", 500)
```

class_data_f2019	<i>class_data_f2019 Data Set</i>
------------------	----------------------------------

Description

This data set contains information on 42 students from Fall of 2019.

Usage

```
class_data_f2019
```

Format

A tibble with 42 rows and 7 variables:

level A factor indicating the class level the student is: Freshman, Sophomore, Junior, Senior, or Graduate.

major A character indicating the major of the student.

sex A factor indicating the sex of the student: F for female or M for male.

ski A factor indicating downhill preference: Ski, Snowboard, or Neither.

penny A factor indicating preference regarding the penny: Abolish, Retain or No Answer.

speed An integer indicating the fastest speed the student had driven in a vehicle (in mph).

sleep A numeric variable indicating how long the student slept the night before.

diseases

diseases Data Set

Description

This data set contains information for each state and Washington, D.C. about the number of reported cases of AIDS, syphilis, and tuberculosis.

Usage

diseases

Format

A tibble with 51 rows and 4 variables:

State A character vector indicating the state.

AIDS The number of reported AIDS cases.

Syphilis The number of reported syphilis cases.

Tuberculosis The number of reported tuberculosis cases.

epilepsy

epilepsy Data Set

Description

This data set contains information on the number of seizures, which treatment, and the age of 59 patients with epilepsy.

Usage

epilepsy

Format

A tibble with 59 rows and 4 variables:

id The patient ID.

numseiz The number of seizures the patient had.

age The age of the patient

 idealwt

idealwt Data Set

Description

This data set contains weight information on 182 people (119 females and 63 males). Actual weights, ideal weights, and the difference between them are recorded.

Usage

```
idealwt
```

Format

A tibble with 182 rows and 4 variables:

sex A factor indicating the sex of the person: Female or Male.

actual The person's actual weight.

ideal The person's ideal weight.

diff The difference between the person's actual weight and their ideal weight. Negative values indicate that the person weighs less than what they consider ideal.

 ilogit

Compute the Inverse Logit of a Value

Description

A function that computes the inverse logit (the inverse log odds) of a value.

Usage

```
ilogit(x)
```

Arguments

x A value for which that we want to find the inverse.

Examples

```
data(mtcars)
mod <- glm(vs ~ mpg, data = mtcars, family = binomial())
ilogit(predict(mod, data.frame(mpg = 20)))
```

influence_plots	<i>Creates many diagnostic plots</i>
-----------------	--------------------------------------

Description

This function creates many diagnostic plots for a given model. These plots include residual plots, a QQ plot, a leverage plot, a Cook's distance plot, a DfFits plot, and DfBetas plots for the intercept and all slopes.

Usage

```
influence_plots(model, missing_group = NULL)
```

Arguments

model	The model for which we would like these plots. This can be of class "lm" or "glm" with a binomial family.
missing_group	Used for multinomial regression to indicate which group is not being plotted.

Value

This function returns plots. The user must press "enter" or "return" to view subsequent plots.

Examples

```
mod <- lm(mpg ~ disp, data = mtcars)
influence_plots(mod)
```

interaction_plot	<i>Two-way Interaction Plot with ggplot2</i>
------------------	--

Description

This function is similar to [interaction.plot](#), but plots in ggplot2 instead of base R. It plots the mean (or other summary) of the response for two-way combinations of factors, thereby illustrating possible interactions.

Usage

```
interaction_plot(
  df,
  response,
  x_factor,
  group_factor,
  linewidth = 1,
  point_size = 2,
  .f = mean,
  ...
)
```

Arguments

<code>df</code>	The data frame that contains the response and factor variables.
<code>response</code>	The name of the numeric variable giving the response. Can be entered with or without quotes.
<code>x_factor</code>	The name of a factor whose levels will form the x axis. Can be entered with or without quotes.
<code>group_factor</code>	The name of a factor whose levels will split up the other variables. Can be entered with or without quotes.
<code>linewidth</code>	The linewidth of the lines plotted. Enter 0 if the lines should be omitted.
<code>point_size</code>	The size of the points plotted. Enter 0 if the points should be omitted.
<code>...</code>	Other parameters that can be passed to geom_point .

Value

This function returns a two-way interaction plot as a ggplot object. All the ggplot add-on functions can be added onto the object returned for more customization.

Examples

```
data(warpbreaks)
interaction_plot(warpbreaks, breaks, wool, tension)
interaction_plot(warpbreaks, "breaks", "wool", "tension")
```

<code>logistic_plots</code>	<i>Creates many diagnostic plots for logistic or multinomial models</i>
-----------------------------	---

Description

This function feeds into the `influence_plots()` function to create many diagnostic plots for a given logistic or multinomial model. These plots include residual plots, a leverage plot, a Cook's distance plot, a DfFits plot, and DfBetas plots for the intercept and all slopes.

Usage

```
logistic_plots(model)
```

Arguments

<code>model</code>	The model for which we would like these plots. This can be of class "glm" with a binomial family or of class "multinom" from the nnet library.
<code>missing_group</code>	Used for multinomial regression to indicate which group is not being plotted.

Value

This function returns plots. The user must press "enter" or "return" to view subsequent plots.

Examples

```
mod <- glm(vs ~ disp, data = mtcars, family = "binomial")
logistic_plots(mod)
```

mlbsalaries

*mlbsalaries Data Set***Description**

This data set contains information on 877 MLB players from 2018.

Usage

```
mlbsalaries
```

Format

A tibble with 877 rows and 5 variables:

Rank The ranking of the player's salary with 1 being the most money earned.

Name A character vector containing the name of the player.

Team A character vector containing the team the player played for.

Position A factor indicating what position the player played: SP for starting pitcher, 1B for first base, 2B for second base, 3B for third base, SS for shortstop, OF for outfield, RP for relief pitcher, and DH for designated hitter.

Salary A numeric vector containing the salary the player made for the 2018 season.

multinom_summary

Outputs estimates, standard errors, test statistic, and p-values for a multinomial logistic regression object from the nnet library.

Description

When given an nnet multinom object, this function will compute the test statistic and p-value for each term in the models and return a tibble with the coefficients, standard errors, test statistic, and p-values in a list with one list element per model.

Usage

```
multinom_summary(model)
```

Arguments

`model` An object of type "multinom" "nnet"

Value

This function returns a list of tibbles with the coefficient estimates, the standard errors, the test statistics, and the p-values for the models comparing each category to the baseline. The AIC and deviance are also given as the last elements in the list.

Examples

```
mod <- mtcars |>
  mutate(carb = factor(carb)) |>
  multinom(carb ~ mpg + disp + hp, data = _)
multinom_summary(mod)
```

nitrogen	<i>nitrogen Data Set</i>
----------	--------------------------

Description

Nitrogen content of trees in an orchard, the growing tips of 150 leaves are clipped from trees throughout the orchard. These leaves are ground to form one composite sample, which the researcher assays for percentage of nitrogen. Composite samples obtained from a random sample of 36 orchards throughout the state gave the nitrogen contents.

Usage

```
nitrogen
```

Format

A data frame with 36 rows 1 variable:

nitrogen The nitrogen content for the trees.

norm_test	<i>Performs either the Shapiro-Francia or the Shapiro-Wilk normality test</i>
-----------	---

Description

When given a model, this will perform either the Shapiro-Francia or the Shapiro-Wilk normality test depending on user request or based on sample size. Both of these tests have a null hypothesis that the residuals are normally distributed and an alternative that the residuals are not normally distributed.

Usage

```
norm_test(model, resid_type = "raw", test = "default", plot_graph = TRUE)
```

Arguments

model	A model of type lm or glm.
resid_type	The type of residuals. The default is "raw", but supports "jackknife", and "standard" as well.
test	The type of test that should be performed. Options are "default", "sf", and "sw". "sf" performs the Shapiro-Francia test, "sw" performs the Shapiro-Wilk, and "default" performs the Shapiro-Francia if $n > 30$ or performs the Shapiro-Wilk if $n \leq 30$.
plot_graph	Whether to return a plot or not. The default is TRUE.

Value

This function returns a test statistic, and p-value and, if requested a plot of the residuals

Examples

```
mod <- lm(mpg ~ disp, data = mtcars)
bf_test(mod)
```

plot.boot3150	<i>Plots a boot3150 object</i>
---------------	--------------------------------

Description

When given a boot3150 object, this function plots the samples with both histograms and QQ plots.

Usage

```
## S3 method for class 'boot3150'
plot(x)
```

Arguments

x An object of class boot3150

Value

This function plots the boot3150 object.

plot_tukey	<i>Plots the Tukey HSD Confidence Intervals</i>
------------	---

Description

This function plots the Tukey HSD confidence intervals when given a tibble outputted by the [tukeyhsd](#) function. If more than one variable, all will be plotted by default, but that can be adjusted with the which option.

Usage

```
plot_tukey(t, which = "all")
```

Arguments

t The Tukey HSD tibble outputted by the [tukeyhsd](#) function.

which Indicates which variable for which the graph should be created. The default is "all", but a variable name used in the model can also be entered (in quotes).

Value

This function returns plots. The user must press "enter" or "return" to view subsequent plots if there are any.

Examples

```
data(warpbreaks)
fm1 <- aov(breaks ~ wool + tension, data = warpbreaks)
t <- tukeyhsd(fm1, "tension", ordered = TRUE)
plot_tukey(t)
```

print.boot3150	<i>Prints a boot3150 object</i>
----------------	---------------------------------

Description

When given a boot3150 object, this function ensures it prints correctly.

Usage

```
## S3 method for class 'boot3150'
print(x)
```

Arguments

x An object of class boot3150

Value

This function prints the output for a boot3150 object.

print.boot3150ci	<i>Prints a boot3150ci object</i>
------------------	-----------------------------------

Description

When given a boot3150ci object, this function ensures it prints correctly.

Usage

```
## S3 method for class 'boot3150ci'
print(x)
```

Arguments

x An object of class boot3150ci

Value

This function prints the output for a boot3150ci object.

```
print.innerboot3150ci Prints a innerboot3150ci object
```

Description

When given a innerboot3150ci object, this function ensures it prints correctly.

Usage

```
## S3 method for class 'innerboot3150ci'
print(x)
```

Arguments

x An object of class innerboot3150ci

Value

This function prints the output for a innerboot3150ci object.

```
residual_plots              Creates residual and QQ plots
```

Description

This function creates a residual and a QQ plot for a given model.

Usage

```
residual_plots(model, resid_type = "jackknife", smoother = TRUE, se = T)
```

Arguments

model	The model for which we would like these plots. This can be of class "lm" or "glm" with a binomial family.
resid_type	The type of residuals that should be used in the plots. The options are "jackknife", "raw", "standard", and "pearson". Jackknife (externally studentized) residuals are used by default for "lm" objects. Standardized Pearson residuals are used for "glm" objects regardless of what is entered here.
smoother	Logical indicating whether a loess smoother should be added to the residual plot.
se	Logical indicating whether the standard error bands should be added to the smoother.

Value

This function returns plots. The user must press "enter" or "return" to view subsequent plots.

Examples

```
mod <- lm(mpg ~ disp, data = mtcars)
residual_plots(mod)
```

test_levene	<i>Performs Levene's test many times to investigate standard deviations needed to reject that the true SDs are equal</i>
-------------	--

Description

This function performs Leven's test for the equality of variance in the ANOVA setting many times for three randomly generated normal samples when given the sample size and records how often the null is rejected, what the standard deviations for each group is when the null is rejected, the p-values when the null was rejected, and the ratio of largest SD to smallest SD for each time the null was rejected.

Usage

```
test_levene(n, num_sims)
```

Arguments

n	Sample size of each group.
num_sims	The number of Levene's tests that should be performed.

Value

This function returns a list that contain the standard deviations for each group when we reject the null hypothesis in Levene's test, the p-values for each time we rejected the null hypothesis, the number of samples, the ratio of largest SD to smallest SD for each time we rejected the null, and the number of times we rejected the null.

Examples

```
test_levene(10, 1000)
```

tukeyhsd	<i>Compute Tukey Honest Significant Differences</i>
----------	---

Description

A tidy version of [TukeyHSD](#). Create a set of confidence intervals on the differences between the means of the levels of a factor with the specified family-wise probability of coverage. The intervals are based on the Studentized range statistic, Tukey's 'Honest Significant Difference' method.

Usage

```
tukeyhsd(model, which, ordered = FALSE, conf.level = 0.95, ...)
```

Arguments

<code>model</code>	A fitted model object, usually an aov fit.
<code>which</code>	A character vector listing terms in the fitted model for which the intervals should be calculated. Defaults to all the terms.
<code>ordered</code>	value indicating if the levels of the factor should be ordered according to increasing average in the sample before taking differences. If <code>ordered</code> is <code>true</code> then the calculated differences in the means will all be positive. The significant differences will be those for which the lower interval end point is positive.
<code>conf.level</code>	A numeric value between zero and one giving the family-wise confidence level to use.
<code>...</code>	Optional additional arguments. None are used at present.

Examples

```
data(warpbreaks)
summary(fm1 <- aov(breaks ~ wool + tension, data = warpbreaks))
tukeyhsd(fm1, "tension", ordered = TRUE)
```

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